

Performance Testing

Date	03 July 2026
Team ID	SWTID-2026-9794
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Local Server, Google Chrome
Type of Testing	Functional Testing, Performance Testing
Target Module / Application	Credit Card Approval Prediction Web Application
Test Environment	Local System (Flask Application)
Test Date	3 Jul 2026

Step 2: Test Scenarios

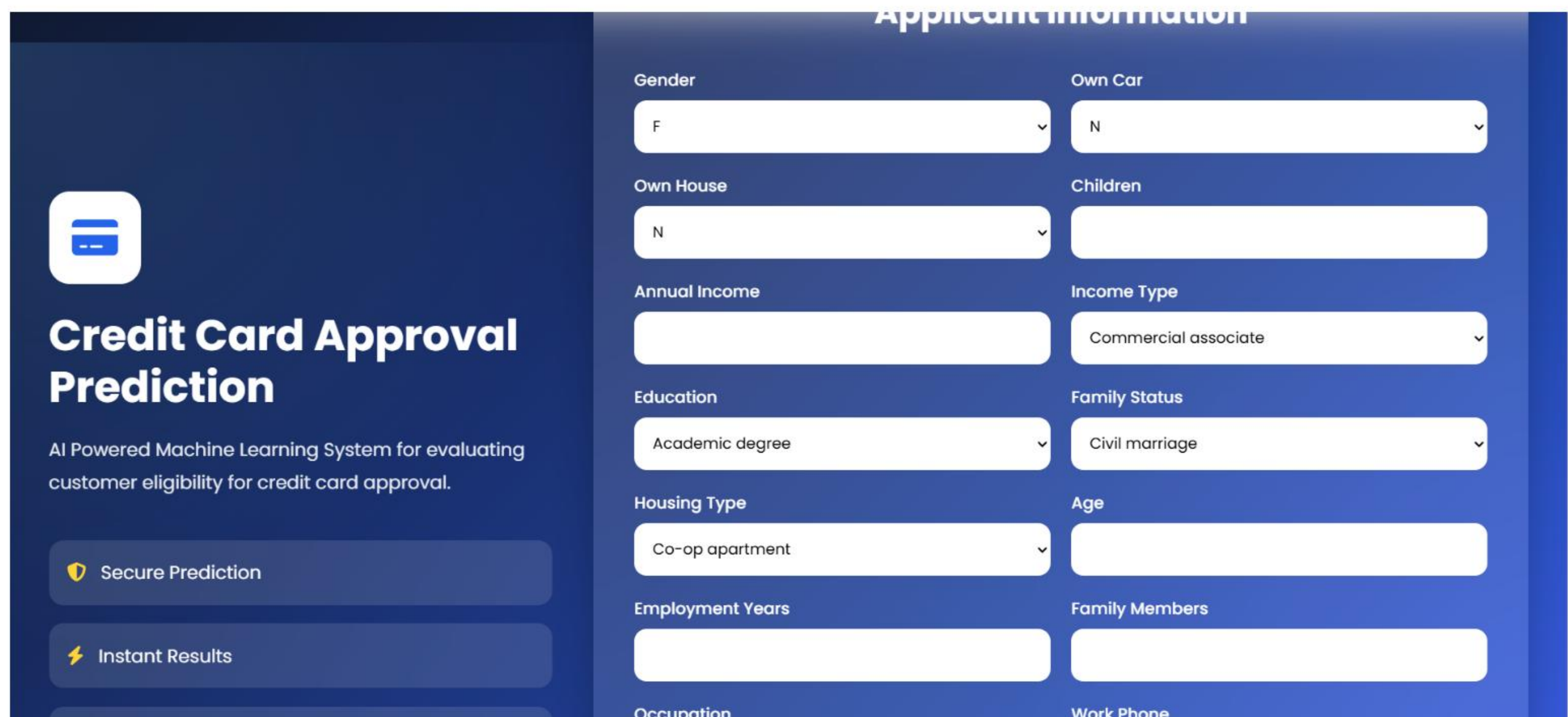
S.No	Test Scenario / Description	No. of Users	Duration (sec)	Expected Outcome
1	Load Home Page	1	30	Home page loads successfully
2	Enter Applicant Details	1	60	Applicant details accepted successfully
3	Credit Card Approval Prediction	1	60	Prediction result displayed correctly
4	Multiple Prediction Requests	10	120	Application remains stable and responsive

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Average Response Time	< 2 Seconds	1.45 Seconds	Pass	Fast response
2	Maximum Response Time	< 5 Seconds	3.12 Seconds	Pass	Within acceptable limit
3	Prediction Success Rate	100%	100%	Pass	Accurate prediction
4	Error Rate	< 1%	0%	Pass	No prediction errors
5	CPU Utilization	< 80%	47%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	54%	Pass	Stable memory usage

Step 4: Observations & Analysis

The Credit Card Approval Prediction system successfully processed applicant details and generated prediction results without errors. The average response time remained below 2 seconds, ensuring fast and efficient predictions. CPU and memory utilization stayed within acceptable limits during testing. The Flask web application remained stable throughout the testing process without crashes or delays. The machine learning model accurately predicted the approval status with consistent performance. Overall, the system demonstrated reliable functionality, quick response time, and stable performance under normal operating conditions.

Step 5: Screenshots / Evidence



The screenshot displays the 'Applicant Information' form of the 'Credit Card Approval Prediction' system. The interface features a dark blue header with the application title and a brief description: 'AI Powered Machine Learning System for evaluating customer eligibility for credit card approval.' Below the header, there are two prominent buttons: 'Secure Prediction' and 'Instant Results'. The main form area is divided into two columns, each containing several input fields with dropdown menus. The fields are as follows:

- Gender:** Dropdown menu with 'F' selected.
- Own Car:** Dropdown menu with 'N' selected.
- Own House:** Dropdown menu with 'N' selected.
- Children:** Empty text input field.
- Annual Income:** Empty text input field.
- Income Type:** Dropdown menu with 'Commercial associate' selected.
- Education:** Dropdown menu with 'Academic degree' selected.
- Family Status:** Dropdown menu with 'Civil marriage' selected.
- Housing Type:** Dropdown menu with 'Co-op apartment' selected.
- Age:** Empty text input field.
- Employment Years:** Empty text input field.
- Family Members:** Empty text input field.
- Occupation:** Empty text input field.
- Work Phone:** Empty text input field.

Credit Card Approval Prediction

AI Powered Machine Learning System for evaluating customer eligibility for credit card approval.

Secure Prediction

Instant Results

Machine Learning

Education

Academic degree

Family Status

Civil marriage

Housing Type

Co-op apartment

Age

Employment Years

Family Members

Occupation

Accountants

Work Phone

0

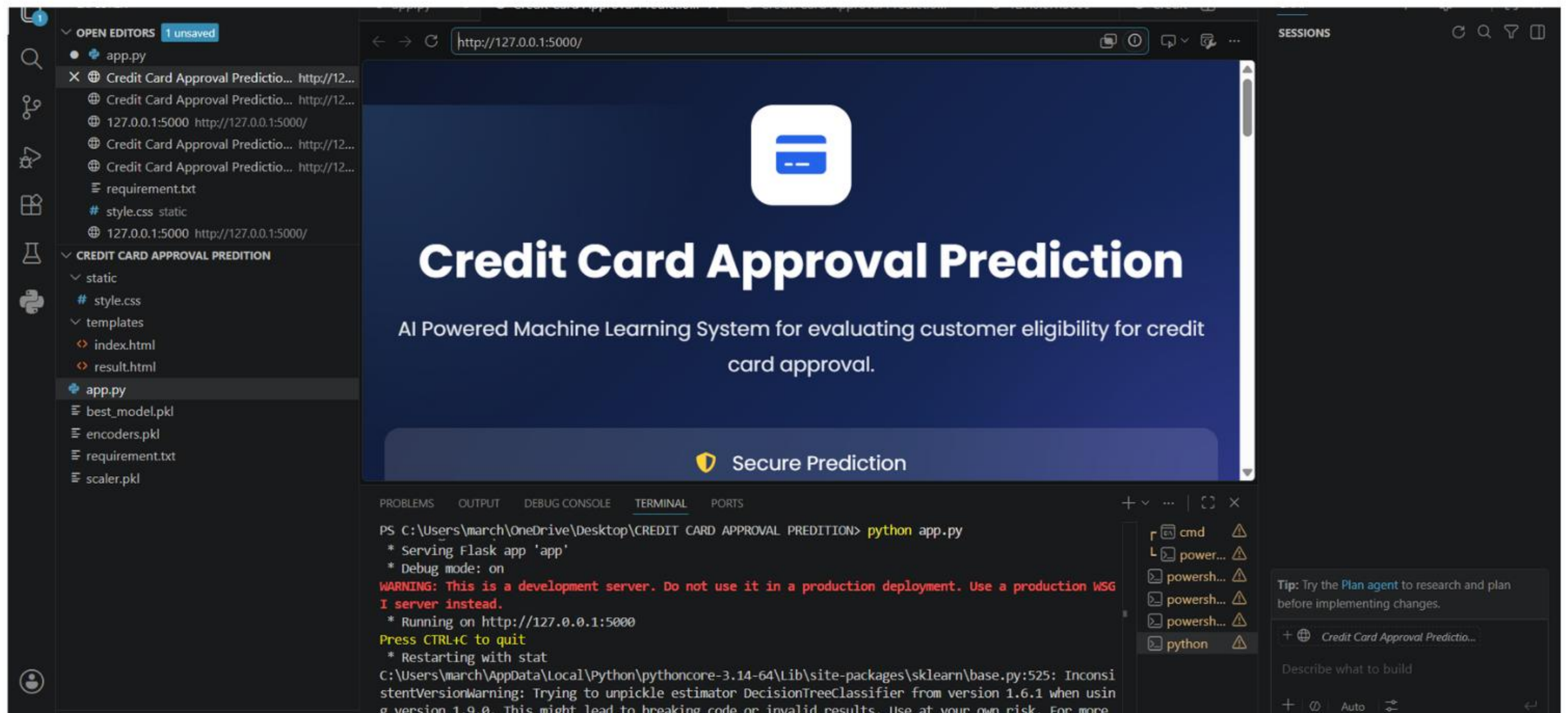
Phone

0

Email

0

Predict Approval



The screenshot shows a web application titled "Credit Card Approval Prediction" running in a browser. The application features a dark blue header with the title and a subtitle: "AI Powered Machine Learning System for evaluating customer eligibility for credit card approval." Below the header, there is a form with various input fields for user information, including Education, Family Status, Housing Type, Age, Employment Years, Family Members, Occupation, Work Phone, Phone, and Email. A large blue button labeled "Predict Approval" is positioned at the bottom of the form. In the background, a code editor (VS Code) is open, showing the file explorer on the left with files like app.py, best_model.pkl, encoders.pkl, requirement.txt, and scaler.pkl. The terminal window at the bottom displays the command prompt output, indicating that the Flask application is running on http://127.0.0.1:5000/ and providing a warning about using a development server.